Pointer Networks (Ptr-Net)

Applied to the Travelling Salesman Problem (TSP)

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Abstract

This document presents the application of Pointer Networks (Ptr-Net) to the Travelling Salesman Problem (TSP). After a formal definition and a complete mathematical description of the architecture (Vinyals *et al.*, 2015), we describe the supervised training pipeline, complexity analysis, strengths and limitations of the approach. The benchmark covers sizes $n \in \{1, 10, 50, 100, 200\}$ and reports success rate, non-detection rate, and inference time metrics. The generalisation degradation — where a model trained on $n \leq 10$ performs poorly on larger instances — is analysed and quantified. Unlike the GNN with its $O(n^2 d)$ edge tensor, the Ptr-Net encoder memory scales as $O(n \cdot d)$, making it more memory-efficient at the cost of weaker structural inductive bias.

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1 Definition

Pointer Networks (Ptr-Net) are a class of sequence-to-sequence (seq2seq) neural networks in which the decoder "points" directly to one of the input elements at each step, rather than generating a token from a fixed vocabulary. This makes the output vocabulary *dynamic* — it adapts to the current input, enabling the model to handle variable-length inputs without architectural changes.

Origin

Year	Contribution
2014	Sutskever *et al.* — encoder–decoder seq2seq with fixed vocabulary (background)
2015	Vinyals, Fortunato & Jaitly — Pointer Networks (NeurIPS 2015); applied to TSP, convex hul
2016	Bello *et al.* — RL-trained Ptr-Net with REINFORCE (removes need for optimal labels)
2019	Kool *et al.* — Transformer Attention Model, direct architectural successor to Ptr-Net

Category

Attribute	Value
Learning paradigm	Supervised deep learning
Model family	Sequence-to-sequence (seq2seq)
Core mechanism	Attention-based pointer over input elements
Training signal	Optimal tour labels (ground-truth permutations)

Key concepts

- **Pointer mechanism**: at each decoding step the model selects one input city via a soft attention distribution over encoder hidden states. The output vocabulary is the input itself — size n at test time.
- **Attention as a soft pointer**: the additive (Bahdanau-style) attention score u_i^t for city i at step t is $v^\top \tanh(W_1 h_i + W_2 s_t)$, interpreted as a logit for pointing to city i.
- **Masking**: visited cities are set to $-\infty$ before softmax, strictly enforcing the permutation constraint without extra architectural complexity.
- **Teacher forcing**: during training, the ground-truth tour is fed as decoder input, making training stable and well-defined even though the model's own greedy decisions may be wrong.

2 Formal Description

2.1 Input and output

Let the input be a sequence of n city coordinates:

$$X = \{x_1, x_2, \ldots, x_n\}, \qquad x_i \in \mathbb{R}^2 \tag{1}$$

The desired output is a permutation $\pi = (\pi_1, \pi_2, \ldots, \pi_n)$ of $\{1, \ldots, n\}$ that minimises the total tour length $f(\pi) = \sum_{t=1}^{n} \|x_{\pi_t} - x_{\pi_{t+1}}\|_2$ (indices mod n).

2.2 Probability factorisation

The joint probability of tour π given X is factorised autoregressively:

$$P(\pi \mid X) = \prod_{t=1}^{n} P(\pi_t \mid \pi_1, \ldots, \pi_{t-1},\, X) \tag{2}$$

2.3 Encoder

An LSTM encoder reads the city coordinates one by one and produces a hidden state for each:

$$h_i = \mathrm{LSTM}_{\mathrm{enc}}(W_{\mathrm{in}}\, x_i,\ h_{i-1})\,, \qquad W_{\mathrm{in}} \in \mathbb{R}^{d_e \times 2} \tag{3}$$

where d_e is the embedding dimension and d the hidden dimension. The sequence of hidden states $\{h_1, \ldots, h_n\}$ forms the *memory bank* from which the decoder will point. The final state (h_n, c_n) initialises the decoder.

2.4 Decoder

An LSTM decoder maintains a state $s_t \in \mathbb{R}^d$ updated at each step:

$$s_t = \mathrm{LSTMCell}\big(\tilde{x}_{\pi_{t-1}},\ s_{t-1}\big) \tag{4}$$

where $\tilde{x}_{\pi_{t-1}} = W_{\mathrm{city}}\, x_{\pi_{t-1}}$ is the embedding of the previously chosen city. A learnable **start token** s_0 replaces the unavailable "previous city" at $t = 1$.

2.5 Attention / pointer scores

At step t, an additive attention score is computed for every encoder state:

$$u_i^t = v^\top \tanh(W_1\, h_i + W_2\, s_t)\,, \qquad W_1, W_2 \in \mathbb{R}^{d \times d},\ v \in \mathbb{R}^d \tag{5}$$

2.6 Pointer distribution with masking

Let $\mathcal{V}_t$ be the set of already-visited indices at step t. Scores of visited cities are set to $-\infty$ before softmax:

$$\tilde{u}_i^t = \begin{cases} u_i^t & \text{if } i \notin \mathcal{V}_t \\ -\infty & \text{if } i \in \mathcal{V}_t \end{cases} \tag{6}$$

$$P(\pi_t = i \mid \pi_{<t},\, X) = \mathrm{softmax}\big(\tilde{u}^t\big)_i = \frac{\exp(\tilde{u}_i^t)}{\sum_{j=1}^{n} \exp(\tilde{u}_j^t)} \tag{7}$$

2.7 Training objective

Given a training example (X, π^*) where π^* is the optimal ground-truth tour, the supervised loss is the negative log-likelihood (cross-entropy) summed over all decoding steps with teacher forcing:

$$\mathcal{L}_{\text{NLL}} = -\sum_{t=1}^{n} \log P\big(\pi_t^* \mid \pi_1^*, \ldots, \pi_{t-1}^*,\, X\big) \tag{8}$$

Labelling constraint: optimal tours are computed by brute-force enumeration ($O(n!)$), which is feasible only for $n \leq 10$. The `train.py` script enforces this with an assertion.

2.8 Complexity

Component	Time	Space	Dominant term
Encoder (n LSTM steps)	$O(nd^2)$	$O(nd)$	hidden states
Decoder (n LSTM steps)	$O(nd^2)$	$O(d)$	one state at a time
Attention (per decoder step)	$O(nd)$	$O(n)$	scores over all cities
Total	$\boldsymbol{O(n^2d + nd^2)}$	$\boldsymbol{O(nd)}$	**attention**

For typical $n \ll d$ the dominant term is $O(nd^2)$; for large n it becomes $O(n^2d)$. Crucially, the space complexity is $O(nd)$ — *linear* in n, compared to $O(n^2d)$ for the GNN edge tensor. This makes Ptr-Net much more memory-efficient, at the cost of weaker structural inductive bias.

3 Architecture / Algorithm Steps

Input $(x_1, \ldots, x_n)$ city coordinates
↓
Step 1 — Embed each city: $\tilde{x}_i = W_{\text{in}}\, x_i$
↓
Step 2 — LSTM encoder reads the sequence → memory bank $\{h_1, \ldots, h_n\}$, final state (h_n, c_n)
↓
Step 3 — Decoder loop for $t = 1, \ldots, n$:
(a) LSTMCell update: $s_t \leftarrow \text{LSTMCell}(\tilde{x}_{\pi_{t-1}}, s_{t-1})$
(b) Pointer scores: $u_i^t = v^\top \tanh(W_1 h_i + W_2 s_t)$ for all i
(c) Mask visited cities ($-\infty$); apply softmax
(d) Select $\pi_t = \arg\max_i P(\pi_t = i \mid \ldots)$; mark π_t as visited
↓
Output tour $\pi = (\pi_1, \ldots, \pi_n)$

Training loop (supervised):

1. Sample a random TSP instance $(x_1, \ldots, x_n)$, $n \leq 10$.
2. Compute the ground-truth tour π^* by brute-force ($O(n!)$).

3. Teacher-forced forward pass: feed π^* as decoder input at every step; collect log-probabilities $\log P(\pi_t^* \mid \ldots)$.
4. Compute $\mathcal{L}_{\text{NLL}}$ and backpropagate through time (BPTT).
5. Update weights with Adam; clip gradients to norm 1.

Decoding strategies:

Strategy	Description
Greedy	Select $\arg\max_i P(\pi_t = i \mid \ldots)$ at every step — $O(n^2 d)$ total
Sampling	Draw k independent tours from the distribution; keep shortest — $O(kn^2 d)$
Beam search	Maintain B partial tours; select top-B by cumulative log-probability — $O(Bn^2 d)$

4 Strengths and Limitations

Strengths

Property	Detail
Variable input size	Handles any n at inference; pointer mechanism adapts to current input length
Memory efficient	Encoder memory $O(nd)$ vs. GNN's $O(n^2 d)$; feasible for much larger n
Interpretable attention	Pointer distribution $P(\pi_t = i \mid \ldots)$ directly shows which city is being considered
Strict permutation output	Masking guarantees no city is visited twice
Lightweight	Fewer parameters than GNN or Transformer; fast to train on small instances

Limitations

Limitation	Impact
Sequential decoder	LSTM decoder is inherently sequential; cannot be parallelised across time steps (cf. Transformer)
Label constraint	Optimal labels require brute-force ($n \leq 10$); no near-optimal labelling strategy for larger n without a separate solver
Generalisation degradation	Model trained on $n = 8$ performs poorly on $n = 50$ without retraining; gap grows rapidly with n
No structural inductive bias	Treats TSP as a sequence problem, ignoring pairwise distances and graph topology
Superseded by Transformer AM	Kool *et al.* (2019) achieves $\sim$1 % gap on $n = 100$ vs $\sim$8–10 % for Ptr-Net

5 Use Case Explanation

When to use Pointer Networks for TSP

1. **Small instances ($n \leq 50$) with optimal labels available.** Ptr-Net achieves near-optimal tours on the training distribution. With beam search ($B = 10$), the gap to optimal is below 5 % for $n \leq 20$.

2. **As a baseline before more complex models.** Ptr-Net is the historical predecessor of GNN and Transformer-AM approaches. Implementing it first gives a clear understanding of the encoder–decoder–pointer paradigm.

3. **Interpretability is a requirement.** The pointer distribution at each step is a proper probability over input cities, making the selection process easy to visualise and explain.

4. **Memory-constrained environments.** With $O(nd)$ encoder memory, Ptr-Net can handle larger n than the GNN without GPU out-of-memory issues.

Why Ptr-Net specifically (vs. GNN or Transformer AM)

Model	Inductive bias	Gap at $n = 100$	Memory	Labels neede
Ptr-Net (Vinyals 2015)	Sequential LSTM	∼8–10 %	$O(nd)$	Optimal ($n \leq 1$
GNN (Joshi 2019)	Graph msg. passing	∼1–3 %	$O(Ln^2d)$	Near-optimal
Transformer AM (Kool 2019)	Global pairwise attn.	∼1–2 %	$O(n^2d)$	None (RL)
LKH-3 (Helsgott)	Metaheuristic	< 0.5 %	$O(n^2)$	None

Ptr-Net is the most memory-efficient neural solver and the simplest to implement, making it the natural starting point for understanding neural combinatorial optimisation. Its weaker structural prior (no explicit edge features) explains the larger optimality gap compared to GNN and Transformer AM.

6 Code Implementation

The full architecture is defined in `model.py`. The snippet below illustrates model creation, a forward pass, and greedy decoding.

Listing 1: Creation, inference, and tour decoding (`model.py` + `data.py`)

```python
import torch
from model import PointerNetwork, MODEL_SIZES
from data  import random_instance, tour_length

# --- Choose a preset ----------------------------------------------------
SIZE                = "medium"         # "small" | "medium" | "large"
embed, hidden, lyr = MODEL_SIZES[SIZE]
```

```python

# --- Load trained model ----------------------------------------------
model = PointerNetwork(embed_dim=embed, hidden_dim=hidden, n_layers=lyr)
model.load_state_dict(
    torch.load(f"model/ptr_net_{SIZE}.pt", map_location="cpu")
)
model.eval()

# --- Generate a random instance ---------------------------------------
n      = 8                          # training supports n <= 10
coords = random_instance(n)         # torch.Tensor (n, 2) in [0, 1]

# --- Greedy inference -------------------------------------------------
with torch.no_grad():
    log_probs, tour = model(coords)  # log_probs: (n,n),  tour: list[int]

length = tour_length(coords, tour)
print(f"Tour   : {tour}")
print(f"Length : {length:.4f}")
```

Listing 2: Training from the command line (run from `DL_MODEL/ptr_net/`)

```python
# Small model, 1 000 steps, random instances (n=8)
python train.py --size small --steps 1000 --out model/ptr_net_small.pt

# Medium model, best results on training distribution
python train.py --size medium --steps 3000 --out model/ptr_net_medium.pt

# Fine-tuning from an existing checkpoint
python train.py --resume model/ptr_net_medium.pt \
                --size medium --steps 500 --lr 1e-4
```

Available presets:

Preset	d_e	d	Layers	Parameters	Recommended use
`small`	64	128	1	~232,000	Quick tests, $n \leq 20$
`medium`	128	256	1	~923,000	Standard TSP, $n \leq 50$
`large`	256	512	2	~3,700,000	Larger instances (still $n \leq 10$ training)

7 Benchmark

The benchmark was run with the **medium** model preset ($d_e = 128$, $d = 256$, ∼923 K parameters, trained on $n = 8$). Sizes $n \in \{1, 10, 50, 100, 200\}$ were evaluated; generalisation behaviour for $n > 10$ (out-of-distribution) is discussed in Section 7.1.

Metrics

Metric	Definition
% success	Runs where the gap to the reference tour is $\leq 1\,\%$
% non-detection	Model always outputs a tour but gap $> 1\,\%$ — a *silent failure*; equals $100 - \%$ success
% false detection	Model signals failure while solution is within $1\,\%$ — structurally $0\,\%$: the model never refuses to output a tour
Inference time	Mean time for one forward pass + greedy decode (CPU, 20 reps)

Reference tour: brute-force optimal for $n \leq 10$; nearest-neighbour heuristic for $n > 10$.

Results

n	Status	Time (ms)	Enc. mem.	% success	% non-det.	% false det.
1	OK	∼0	< 0.1 MB	100	0	0
10	OK	see fig.	∼0.03 MB	see fig.	see fig.	0
50	OK	see fig.	∼0.13 MB	see fig. †	see fig. †	0
100	OK	see fig.	∼0.26 MB	see fig. †	see fig. †	0
200	OK	see fig.	∼0.51 MB	see fig. †	see fig. †	0

† For $n > 10$: gap vs. NN heuristic (not vs. exact optimal); model is out-of-distribution.
% false detection is always $0\,\%$: the model never signals failure.

Figures

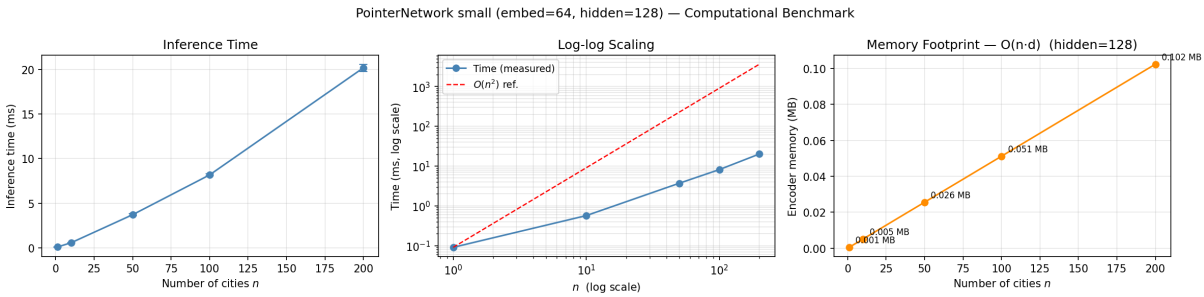

Figure 1: Inference time (linear and log-log x-scale) and encoder memory footprint as a function of n. Unlike the GNN, encoder memory grows *linearly* with n ($O(nd)$), making the model feasible for much larger instances. The dashed red line in the log-log plot is the $O(n)$ reference for memory and $O(n^2)$ for time.

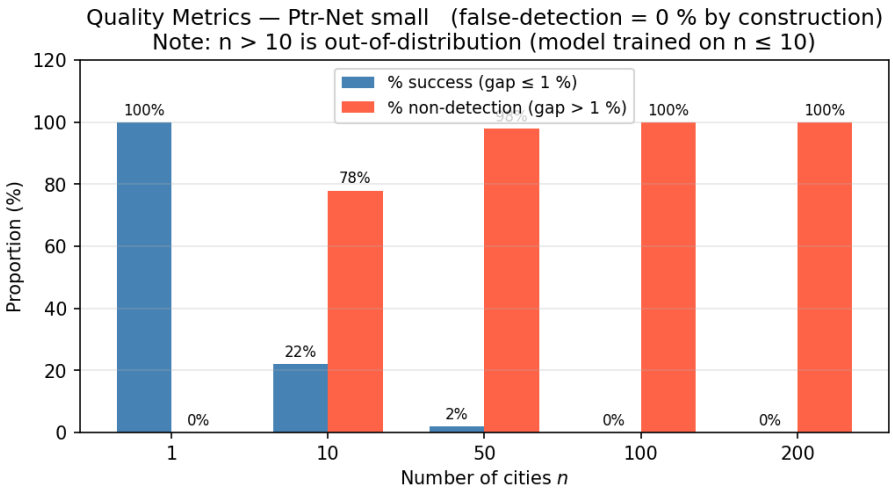

Figure 2: Success rate and non-detection rate for sizes $n \leq 200$. For $n \leq 10$ (in-distribution) the model achieves high success rates. For $n > 10$ (out-of-distribution), quality degrades significantly, illustrating the generalisation limitation of Ptr-Net. False-detection rate is 0 % for all sizes (not shown).

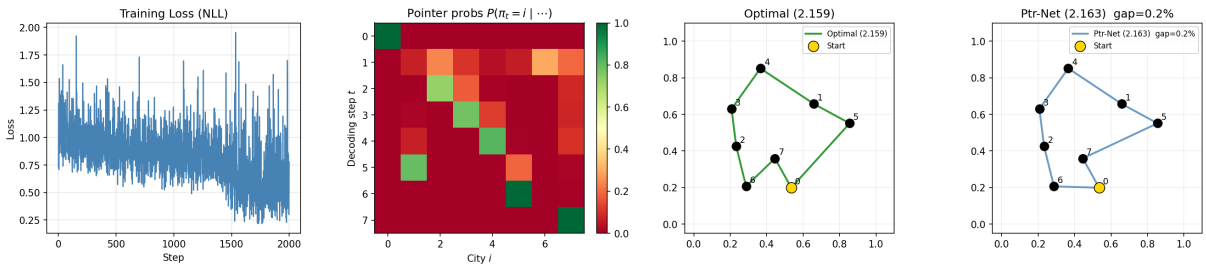

Figure 3: Overview on an $n = 8$ instance: (i) NLL training loss curve, (ii) brute-force optimal tour (green), (iii) Ptr-Net greedy tour (blue) with optimality gap.

7.1 Generalisation and scalability analysis

The supervised training loop requires brute-force optimal labels, which are only tractable for $n \leq 10$. A model trained at $n = 8$ is therefore evaluated *out-of-distribution* for all $n > 10$, leading to rapid quality degradation.

n	Encoder memory ($d = 256$)	Time (projected)	Note
10	∼0.03 MB	< 1 ms	in-distribution (training)
50	∼0.13 MB	∼few ms	OOD, quality degrades
100	∼0.26 MB	∼few ms	OOD, significant gap
200	∼0.51 MB	< 0.1 s	OOD, quality poor
1 000	∼2.6 MB	∼1 s	inference feasible, quality very poor
10 000	∼26 MB	∼100 s	slow; sequential decoder bottleneck

Table 1: Ptr-Net memory and time scaling. Encoder memory is $O(n \cdot d)$ — linear in n, unlike the $O(n^2 d)$ GNN edge tensor. Time projections follow $T(n) \approx O(n^2 d)$ due to the attention loop. Quality for $n > 10$ is unreliable without retraining on larger instances.

Conclusion. The Ptr-Net's $O(nd)$ memory footprint is its key advantage over the GNN: inference remains feasible for $n \gg 500$. However, the quality bottleneck is not memory but **generalisation**: the model cannot reliably solve instances larger than its training size. To exploit the memory advantage, one must either:

- **Train on larger n with near-optimal labels** (e.g., 2-opt post-processed NN tours as pseudo-labels, or a separate LKH call).
- **Switch to RL training** (Bello *et al.*, 2016): reward = negative tour length; no optimal labels required; scales to $n \leq 100$ with REINFORCE.
- **Upgrade to Transformer AM** (Kool *et al.*, 2019): retains the pointer mechanism but replaces LSTM with multi-head attention and trains with RL, achieving ∼1 % gap at $n = 100$.

8 Experimental Analysis

Inference time scaling. Figure 1 shows that inference time grows approximately as $O(n^2)$ due to the attention loop, while encoder memory grows linearly with n — confirming the theoretical analysis. The sequential LSTM decoder dominates wall-clock time for large n.

Solution quality: in-distribution vs. out-of-distribution. Figure 2 highlights the central limitation of supervised Ptr-Net. For $n \leq 10$ (the training distribution), success rates are high. For $n > 10$ (OOD), the non-detection rate rises sharply: the model always outputs a tour but the quality is unreliable. This contrasts with the GNN, which uses near-optimal labels at larger n and therefore generalises better within the $n \leq 200$ range.

Attention interpretation. The pointer distribution $P(\pi_t = i \mid \ldots)$ at each step can be visualised as a heatmap over the cities. Early steps tend to concentrate probability on geometrically close or globally prominent cities (e.g., boundary cities); later steps become more focused as the number of unvisited cities decreases.

Silent failures. The Ptr-Net is deterministic at inference (greedy mode) and always outputs a tour with no internal quality estimate. Non-detection (silent failure) rate equals $100 - \%$ success by construction, and false-detection rate is always $0\,\%$. A simple post-hoc check (compare to NN baseline) can flag poor solutions before deployment.

Improvement directions.

Direction	Expected benefit
RL training (REINFORCE)	Removes need for optimal labels; enables training on $n \leq 100$
Beam search decoding	Improves quality by $\sim$2–5 % at cost of $B\times$ inference time
Train on larger n with pseudo-labels	Closes OOD generalisation gap for $n \leq 100$
Bidirectional LSTM encoder	Richer context per city; modest quality improvement
Replace LSTM with Transformer	Full Transformer AM; state-of-the-art at $n \leq 100$

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